

Fuzzy process control: construction of control charts with fuzzy numbers

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Abstract

This paper presents the construction of fuzzy control charts for a process with fuzzy outcomes derived from the subjective quality ratings provided by a group of experts. The proposed fuzzy process control (FPC) methodology comprises an off-line stage and an on-line stage. In the off-line stage, experts assign quality ratings to products based on a numerical scale. The individual numerical ratings are then aggregated to form collective opinions expressed in the form of fuzzy numbers. The collective knowledge applied by the experts when conducting the quality rating process is acquired through a process of fuzzy regression analysis performed by a neural network. In the on-line stage, the product dimensions are measured, and the fuzzy regression model is employed to automate the experts' judgments by mapping the measured dimensions to appropriate fuzzy quality ratings. The fuzzy quality ratings are then plotted on fuzzy control charts, whose construction and out-of-control conditions are developed using possibility theory. The developed control charts not only monitor the central tendency of the process, but also indicate its degree of fuzziness.

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1. Introduction

Statistical process control (SPC) is a methodology for monitoring a process in order to identify special causes of variation and to signal the requirement for corrective action when necessary. SPC is widely employed throughout industry, and is a proven technique for improving quality and productivity [8].

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SPC relies on the use of control charts to detect whether or not a process is out of control. Typically, a control chart features a centre line (CL) and two control lines, referred to as the upper control limit (UCL) and the lower control limit (LCL), respectively. The centre line represents an estimate of the process level, while the two control limits denote the boundaries of normal variability, and are specified such that the majority of the observations (generally 99.73%) lie within their bounded range when the process is under control. Samples drawn from the process are plotted as points on the control chart. If any of these points fall outside of the control limits, or if any unusual patterns are noted, it is likely that some special cause has affected the process. The control charts are constructed concurrently with the statistical testing process. Essentially, the control chart tests the hypothesis that the process is in a state of statistical control. Accordingly, a plotted data point falling within the control limits confirms the hypothesis of statistical control, while a point falling outside of the control limits indicates a rejection of this hypothesis.

Soft quality control procedures tend to be more frequently applied in production processes where it is difficult to obtain a precise quality description of the inspected item. Under these conditions, the inspector assesses the quality of the inspected item on the basis of his or her subjective judgment, and hence the inspection process yields a vague outcome [11]. In monitoring vague production processes, the vague observations can be reduced to exact numbers, which can then be used to establish statistical control charts. However, this approach has two potential flaws. Firstly, in reducing the vague observations to exact numbers, the information content of the original data is also reduced [11]. Secondly, the uncertainty (i.e., variation) of vague observations is essentially non-statistical in nature, and hence these observations may not adequately support the random variation assumption inherent in statistical quality control methods. Fuzzy set theory specifically addresses the development of concepts and techniques for dealing with sources of uncertainty or imprecision which are non-statistical in nature [31]. Hence, this study proposes a fuzzy process control (FPC) methodology in which fuzzy control charts are employed to monitor a process whose outcomes are represented by fuzzy numbers.

The concepts of fuzzy process control are well documented in the available literature [10–13,19,26]. To overcome the limitations of the binary classification used in control charts for attributes, e.g., the *p*-chart and the *c*-chart, Wang and Raz [26] and Raz and Wang [19] adopted linguistic terms such as *perfect*, *good*, *medium*, *poor*, and *bad* to express the intermediate levels of a quality characteristic rather than the traditional choice of “conforming” or “nonconforming”. In developing control charts for process outcomes based on linguistic terms, the membership functions of the various linguistic terms are defined on a numerical scale of 0–1 (as shown in Fig. 1). Wang and Raz [26] and Raz and Wang [19] proposed two approaches for the construction of control charts, namely a probabilistic approach and a membership approach. In the probabilistic approach, the representative values (crisp real numbers) of the linguistic data were obtained by calculating the modes, medians, or fuzzy averages of their membership functions. These representative values were then utilized to construct control charts using traditional statistical methods. Meanwhile, in the membership approach, the process level was estimated by the average of the total set of linguistic observations, and the representative value of the average was taken as the centre line of the control chart. The control limits were then offset from the centre line by a multiple of the fuzziness quantity of the estimated process level.

Kanagawa et al. [12,13] also adopted linguistic terms to express process outcomes. However, in their case, the control charts were constructed using a different approach to that employed by Wang and Raz [26] or by Raz and Wang [19]. Rather than assuming a probability density function for the representative values of the linguistic terms, Kanagawa et al. [12] considered the linguistic terms as labeled fuzzy data, and calculated the probability of the appearance of a linguistic variable using Zadeh’s probability function

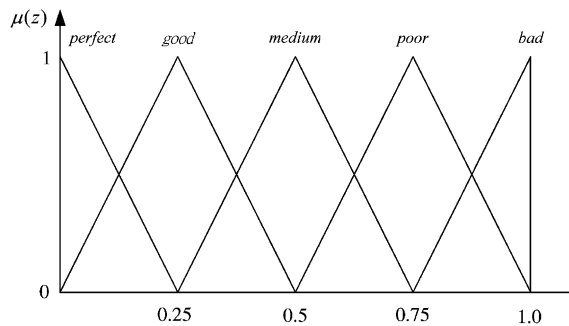


Fig. 1. Linguistic terms used in the control chart of Wang and Raz [26] and Raz and Wang [19].

[30]. The control limits of the control chart were then determined from the probability of points falling outside of the control limits when the process was in control.

Grzegorzewski [10] proposed a fuzzy control chart consisting of two complementary graphs for fuzzy observations represented by fuzzy numbers. The first graph incorporated a centre line which took the representative value of the fuzzy grand mean of all the samples as an estimate of the process level. Each sample was transformed to an interval symbolizing the fuzzy set of the sample mean. These intervals were plotted on the graph, and the failure of an interval to intersect with the centre line was taken as an indication of an out-of-control situation. Each interval also corresponded to a value plotted on the second graph, in which each value was interpreted as a degree of conviction that the process was out of control.

The aforementioned approaches for constructing fuzzy control charts established the centre line by calculating the representative value of the grand sample mean. However, these approaches all have the drawback that the fuzziness of the process level tends to be lost to a certain extent, and hence the resultant control charts lose some of the information associated with the original data. To retain the fuzziness of vague data, Grzegorzewski and Hryniewicz [11] proposed the use of a fuzzy control chart based on a statistical test to verify a fuzzy hypothesis with vague data. Their approach utilized the necessity index of strict dominance (NSD) proposed by Dubois and Prade [5] to test the fuzzy hypothesis and required the user to specify a value for the necessity index, ξ . A centre area, rather than a centre line, was determined by the $(1 - \xi)$ -level set of the grand sample mean, and was used to estimate the process level. The upper and lower control limits were determined using the method presented by Grzegorzewski [10]. Each sample was represented by the interval of the $(1 - \xi)$ -level set of the sample mean. The presence of an interval outside the control limits was taken as an indication that the process was no longer in control.

In general, the aforementioned approaches for fuzzy process control employed the linguistic terms in Fig. 1 to assess product quality. However, the membership functions of these linguistic terms may not accurately reflect the expert's judgment since they are assigned arbitrarily along the scale without regard to the fuzziness of the expert's judgment. Kanagawa et al. [12] also commented that the membership functions of the linguistic terms used in their study and in that of Wang and Raz [26] were problematic.

This study proposes an alternative approach to deal with the experts' subjective judgments. Based on the rating scores assigned by the individual inspectors to the inspected items, fuzzy numbers are constructed to represent the vague outcomes of the process. Fuzzy control charts are then constructed directly from these fuzzy numbers, thereby retaining the fuzziness of the original vague observations. The out-of-control conditions are then formulated using possibility theory.

2. Fuzzy process control procedure

As argued in the previous section, assigning a linguistic term directly to rate the quality of an inspected item is inappropriate when the membership function of the linguistic term is not well constructed. However, it is relatively straightforward for an inspector to assign a numerical score to an inspected item to indicate his or her assessment of its quality. In such an inspection process, different inspectors may assign different scores to the same object. Importantly, this variation is due to a limited perception of the values in the rating scale. To model the limited perception existing in the inspection process, the present study employs fuzzy numbers to aggregate the experts' rating scores to represent the dispersion of the vague observations.

The fuzzy process control method proposed in the present study comprises an off-line stage and an on-line stage. In the former, a sample of products is randomly selected from the manufacturing process, and a small group of experts are invited to grade the quality of each product by assigning an appropriate quality rating score. The various ratings assigned by the individual experts are then aggregated into fuzzy numbers, which are taken to represent the overall quality evaluation of the product. If the proposed FPC methodology is to be efficient, and hence to be of practical value, it is necessary to minimize the involvement of human experts in the online inspection process. This can be achieved by capturing the experts' knowledge via a machine learning process. In this study, the relationship between a product's measurable characteristics (generally its dimensions) and its fuzzy quality ratings is identified by fuzzy regression analysis using a neural network. The resultant fuzzy regression model is then applied on-line to automate the product quality assessment task. In this on-line stage, the product dimensions are measured and the dimensional data are supplied to the fuzzy regression model, which then automatically computes the associated fuzzy quality ratings. In the final step, the fuzzy quality ratings are plotted on control charts, on which the out-of-control conditions are formulated on the basis of possibility and necessity measures.

3. Fuzzy number construction

Let the quality of a product be graded on a scale of 0 to G . Further, let $g_1, \dots, g_n$ be the scores assigned by n different experts when rating the same product. Due to a limited perception of the values in the rating scale, different experts may assign different scores to the same product, and even the same expert may assign a different score when presented with the same product for a second time. The assigned scores can be considered to represent a sample from the possibility distribution of the grading process. Concepts for identifying membership functions from fuzzy observation data have been presented by Tamaki et al. [23]. Their approach assumed that a probability density function lay in the background of the observation, and that its random variable was not observed directly, but rather by way of selecting from a collection of fuzzy sets. However, this condition is not satisfied in the present case since the current observations (i.e. $g_1, \dots, g_n$) are real numbers rather than fuzzy sets. Consequently, this study introduces a novel approach for identifying the membership function of a fuzzy number from a series of observations, $g_1, \dots, g_n$.

Estimating the mode of the fuzzy number involves locating the center around which the g_i 's gather. In the estimation process, scores lying closer to the center are considered to have a greater importance. In other words, the estimation of this center is a weighted average of the g_i 's. To approximate the center of the g_i 's, the relative distances between the various g_i 's are used to locate their congestion. The distances between each g_i are calculated and presented in the relative distance matrix $\mathbf{D} = [d_{ij}]_{n \times n}$,

where $d_{ij} = |g_i - g_j|$, which implies that $d_{ii} = 0$ and $d_{ij} = d_{ji}$. The average of the relative distances for each g_i is calculated by $\bar{d}_i = \sum_{j=1}^n d_{ij} / (n - 1)$. This average distance is then used to measure the proximity of g_i to the center of all the scores. The smaller the value of $\bar{d}_i$, the closer the proximity of g_i to the center and the greater its assigned weight when estimating the mode of the fuzzy number. To determine the degree of importance of each g_i in this estimation method, pair-wise comparisons between g_i 's are conducted based on their average distances. A pair-wise comparison matrix $\mathbf{P} = [p_{ij}]_{n \times n}$, where p_{ij} is the relative importance of g_i compared to g_j , is given by: $p_{ij} = \bar{d}_j / \bar{d}_i$, which implies that $p_{ii} = 1$ and $p_{ij} = 1/p_{ji}$. Since $\mathbf{P}$ is obtained from a comparison of distances, it is perfectly consistent, i.e., there exists a coherent judgment in specifying the pair-wise comparison of the importance of the scores. Let w_i be the true degree of importance of g_i and $0 \leq w_i \leq 1$. Consequently, due to the consistency of $\mathbf{P}$

$$p_{ij} = \frac{w_i}{w_j}, \quad \forall i, j. \quad (1)$$

Let $\mathbf{w}$ be a column vector of w_i , $i = 1, \dots, n$. Therefore, from Eq. (1), $\mathbf{P}\mathbf{w} = n\mathbf{w}$, which implies that n is an eigenvalue of $\mathbf{P}$ and $\mathbf{w}$ is the corresponding eigenvector. The value of $\mathbf{w}$ is solved by the weight determination method of Saaty [20]. Given $\sum_{i=1}^n w_i = 1$, $\mathbf{w}$ is solved as

$$w_j = \frac{1}{\sum_{i=1}^n p_{ij}}, \quad j = 1, \dots, n. \quad (2)$$

The importance degree w_i represents the weight to be associated with g_i when estimating the mode m of the fuzzy number. This estimation is defined as

$$m = \sum_{i=1}^n w_i g_i. \quad (3)$$

Estimating the spreads of a fuzzy number, i.e., determining its left and right endpoints, requires the prior solution of its variance. Carlsson and Fullér [2] defined the possibilistic variance of a fuzzy number F as

$$\begin{aligned} \text{Var}(F) &= \int_0^1 \gamma \left(\frac{e^l(\gamma) + e^u(\gamma)}{2} - e^l(\gamma) \right)^2 d\gamma + \int_0^1 \gamma \left(\frac{e^l(\gamma) + e^u(\gamma)}{2} - e^u(\gamma) \right)^2 d\gamma \\ &= \frac{1}{2} \int_0^1 \gamma [e^u(\gamma) - e^l(\gamma)]^2 d\gamma, \end{aligned}$$

where $e^l(\gamma)$ and $e^u(\gamma)$ are the lower and upper limits of the γ -level set of F , respectively. Fullér and Majlender [9] further generalized the above definition by introducing a weighting function, $\omega(\gamma)$, to measure the importance of the γ -level set of F , and defined the ω -weighted possibilistic variance of F as $\text{Var}_\omega(F) = \int_0^1 \left[\frac{e^u(\gamma) - e^l(\gamma)}{2} \right]^2 \omega(\gamma) d\gamma$, where the weighting function is nonnegative, monotonically increasing and satisfies $\int_0^1 \omega(\gamma) d\gamma = 1$. Note that when $\omega(\gamma) = 2\gamma$, $\text{Var}(F)$ and $\text{Var}_\omega(F)$ are equivalent.

The definition of the ω -weighted possibilistic variance provides the means of using the weights determined by Eq. (2) to estimate the spreads of a fuzzy number. However, to allow the use of the given scores in this estimation, the present study re-defines the ω -weighted possibilistic variance of a triangular fuzzy

number based on the definitions of Carlsson and Fullér [2] and Fullér and Majlender [9], i.e.,

$$\text{Var}_\omega(F) = \frac{1}{2} \left\{ \int_0^1 [m - e^l(\gamma)]^2 \omega(\gamma) d\gamma + \int_0^1 [m - e^u(\gamma)]^2 \omega(\gamma) d\gamma \right\}. \quad (4)$$

In Eq. (4), the mode of the fuzzy number, m , replaces the arithmetic mean of the γ -level set used in the definition of Carlsson and Fullér [2]. When F is symmetric, the ω -weighted possibilistic variance defined in Eq. (4) equals to the variance defined by Fullér and Majlender [9].

Let a and b be the left and right endpoints of F , respectively, and let $\omega(\gamma) = 2\gamma$. Eq. (4) can be rewritten as

$$s^2 = \text{Var}_\omega(F) = \frac{(m - a)^2 + (b - m)^2}{12}. \quad (5)$$

Let the ratio of the left spread to the right spread be expressed by ρ , i.e.,

$$\rho = \frac{m - a}{b - m}. \quad (6)$$

Eqs. (5) and (6) can be solved to yield

$$a = m - \sqrt{\frac{12\rho^2 s^2}{1 + \rho^2}}, \quad (7)$$

and

$$b = m + \sqrt{\frac{12s^2}{1 + \rho^2}}. \quad (8)$$

However, until a and b are solved, s^2 and ρ are unknown. Hence, they must initially be approximated, as shown below.

A variance ($\hat{s}^2$) calculated from the sample of scores is used to approximate s^2 based on Eq. (4), i.e.,

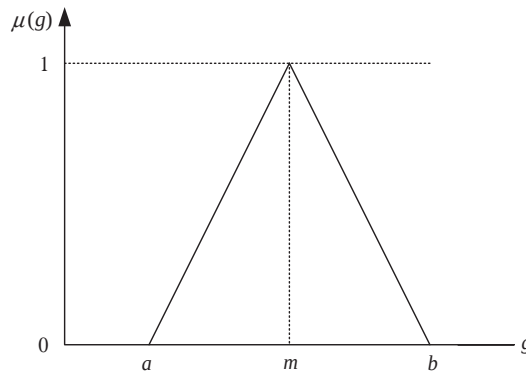
$$\hat{s}^2 = \frac{1}{2} \left\{ \frac{\sum_{i \in \arg\{g_i < m\}} w_i (m - g_i)^2}{\sum_{i \in \arg\{g_i < m\}} w_i} + \frac{\sum_{i \in \arg\{g_i > m\}} w_i (m - g_i)^2}{\sum_{i \in \arg\{g_i > m\}} w_i} \right\}. \quad (9)$$

Regarding the approximation of the ratio ρ , let g^l be the weighted average of the scores which are less than m , and let g^r be the weighted average of the scores which are greater than m . Hence:

$$g^l = \frac{\sum_{i \in \arg\{g_i < m\}} w_i g_i}{\sum_{i \in \arg\{g_i < m\}} w_i}, \quad (10)$$

and

$$g^r = \frac{\sum_{i \in \arg\{g_i > m\}} w_i g_i}{\sum_{i \in \arg\{g_i > m\}} w_i}. \quad (11)$$

Fig. 2. Membership function of a triangular fuzzy number (a, m, b) .

Assuming that g^l and g^r are at the same membership level in F , the ratio of the left spread to the right spread can be approximated as

$$\hat{\rho} = \frac{m - g^l}{g^r - m}. \quad (12)$$

With parameters of m , a and b , the triangular fuzzy number shown in Fig. 2 is defined as

$$t(g; a, m, b) = \begin{cases} 1 - \frac{m - g}{m - a}, & a \leq g \leq m, \\ 1 - \frac{g - m}{b - m}, & m < g \leq b, \\ 0, & \text{elsewhere,} \end{cases} \quad (13)$$

where $0 \leq g \leq G$. Note that a triangular fuzzy number with the form of Eq. (13) is denoted as (a, m, b) .

The construction of a fuzzy number using the proposed method is demonstrated in the following example.

Example 1. It is assumed that five experts inspect the same product. The scores assigned by these experts to the product are given by $g_1 = 4$, $g_2 = 5$, $g_3 = 6$, $g_4 = 6$, and $g_5 = 8$, respectively. The corresponding relative distance matrix is expressed as

$$\mathbf{D} = \begin{bmatrix} 0 & 1 & 2 & 2 & 4 \\ 1 & 0 & 1 & 1 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 & 2 \\ 4 & 3 & 2 & 2 & 0 \end{bmatrix}.$$

The averages of the relative distances for the five scores are

$$\bar{d}_1 = 2.25, \bar{d}_2 = 1.50, \bar{d}_3 = 1.25, \bar{d}_4 = 1.25, \bar{d}_5 = 2.75.$$

The pair-wise comparison matrix is then obtained as

$$\mathbf{P} = \begin{bmatrix} 1.000 & 0.667 & 0.556 & 0.556 & 1.222 \\ 1.500 & 1.000 & 0.833 & 0.833 & 1.833 \\ 1.800 & 1.200 & 1.000 & 1.000 & 2.200 \\ 1.800 & 1.200 & 1.000 & 1.000 & 2.200 \\ 0.818 & 0.545 & 0.455 & 0.455 & 1.000 \end{bmatrix}.$$

From Eq. (2), the weights associated with each of the scores are found to be $w_1 = 0.145$, $w_2 = 0.217$, $w_3 = 0.260$, $w_4 = 0.260$, and $w_5 = 0.118$. The mode of the fuzzy number based on this sample of five scores is estimated from Eq. (3) to be $m = 5.73$. Eq. (9) gives the estimated variance, $\hat{s}^2$, as 1.265, while from Eqs. (10) and (11), it is shown that $g^l = 4.60$ and $g^r = 6.37$, and hence $\hat{\rho} = 0.722$ by Eq. (12). Finally, the left and right endpoints of the fuzzy number are obtained from Eqs. (7) and (8) as $a = 3.45$ and $b = 8.89$, respectively.

Although the present study treats the rating scale as cardinal, ordinal scales are also commonly employed in practice. The additive form of the scores in Eq. (3) may not be appropriate in such a case. When an ordinal scale is used, the statistical mode of the scores may represent a more appropriate option for establishing the mode of the fuzzy number.

4. Fuzzy regression analysis with a neural network

To acquire the knowledge applied by the experts in the quality grading process, this study employs fuzzy regression analysis to identify the relationship between the dimensions of a product and its fuzzy quality rating. Tanaka et al. [25] first proposed the use of fuzzy regression analysis in 1982 to deal with regression problems for vague systems. In such systems, the vagueness is generally a result of human judgment introduced when individuals observe that system.

Having aggregated the experts' scores into fuzzy numbers, the collective knowledge of the experts can be retained by constructing a fuzzy regression model to map the dimensions of the products to their respective fuzzy quality ratings. In the present study, fuzzy regression analysis is performed using the fuzzy radial basis function network (FRBFN) proposed by Cheng and Lee [3]. The architecture of the FRBFN is depicted in Fig. 3.

The input layer of the FRBFN captures the dimensions of the product, while the output node produces a fuzzy number, $\hat{Y} = (\hat{a}, \hat{m}, \hat{b})$, which simulates the collective quality rating for the product of the specified dimensions. As shown in Fig. 3, the FRBFN has a single hidden layer with q hidden units, where the active functions h_j , $j = 1, \dots, q$, of these hidden units are Gaussian functions, i.e.,

$$h_j(\mathbf{x}) = \exp \left[- \left(\frac{\mathbf{x} - \mathbf{u}_j}{\sigma_j} \right)^2 \right], \quad (14)$$

where $\mathbf{x} = [x_1, \dots, x_p]^T$ is the p -dimensional input vector, and $\mathbf{u}_j = [u_1, \dots, u_p]^T$ and σ_j are the center and deviation of the Gaussian function, respectively. The arrangements of $\mathbf{u}_j$ and σ_j are such that appropriate clusters are formed from the input vectors in order that the hidden units cover the portions of input space within which the training data actually occur [22]. The connection weights $v_j =$

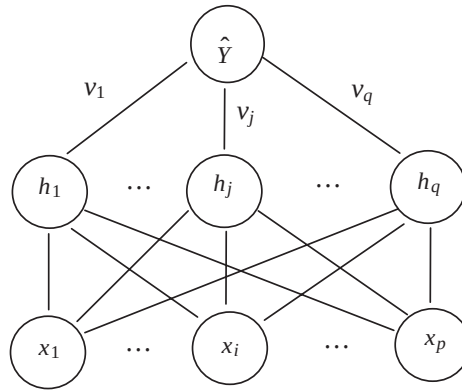


Fig. 3. Architecture of FRBFN [3].

(l_j, c_j, r_j) , $j = 1, \dots, q$, between the hidden layer and the output node are triangular fuzzy numbers of the form given in Eq. (13). Each connection weight can be interpreted as the desired output for an input vector located exactly at the center of its corresponding hidden unit. The output of the FRBFN, $\hat{Y}$, is computed through fuzzy arithmetic, and is given by

$$\hat{Y} = \frac{\sum_{j=1}^q h_j(\mathbf{x}) v_j}{\sum_{j=1}^q h_j(\mathbf{x})}. \quad (15)$$

4.1. Learning algorithms for FRBFN

A hybrid-learning algorithm is formulated for the FRBFN in which the learning of the parameters in the hidden units is self-organized while that of the connection weights is supervised. The aim of the hidden unit parameter learning task is to arrange the hidden units such that they can effectively represent the clusters of input vectors. The c -mean clustering algorithm is used to achieve this goal. The resultant updating rule for $\mathbf{u}_j$ is expressed as

$$\mathbf{u}_j^{\text{new}} = \mathbf{u}_j^{\text{old}} + \eta(\mathbf{x} - \mathbf{u}_j^{\text{old}}), \quad (16)$$

where η is the learning rate.

The initial values of the $\mathbf{u}_j$'s are roughly determined by observing the dispersion of the sample data. The learning of σ_j is determined using the k -nearest neighbour heuristics. This heuristic scheme varies the value of σ_j in order to achieve a certain overlap between the hidden units such that they form a smooth and continuous interpolation over the input space. σ_j is updated by taking the average of the Euclidean distances between the j th unit center and all of its k -nearest neighbouring unit centers.

Eq. (15) indicates that the output of the FRBFN is derived from a fuzzy linear equation of the hidden units' outputs. This implies that the learning of the connection weights can be formulated as a fuzzy linear regression problem. From Eq. (15), it can be seen that when the $h_j(\mathbf{x})$'s are fixed, $\hat{Y}$ is a possibilistic linear equation incorporating v_j 's as unknown fuzzy parameters. The identification of a possibilistic linear equation is widely discussed in the available literature [1,7,14,17,21,27,28]. However, the approach

employed in the present study is based mainly on the proposals of Tanaka [24] and Lee and Tanaka [16] as described in the following paragraph.

It is assumed that N objects are collected and that their dimensions are measured as $\mathbf{x}_1, \dots, \mathbf{x}_N$, respectively. The quality of each object is then evaluated by experts and the corresponding ratings are converted into N fuzzy numbers, i.e., $Y_1, \dots, Y_N$, as discussed in the previous section, where $Y_k = (a_k, m_k, b_k)$, $k = 1, \dots, N$. Meanwhile, let the network outputs for the given $\mathbf{x}_k$'s be denoted as $\hat{Y}_k = (\hat{a}_k, \hat{m}_k, \hat{b}_k)$, where $k = 1, \dots, N$. From Eq. (15), and through fuzzy arithmetic, it is determined that $\hat{a}_k = \sum_{j=1}^q f_{jk} l_j$, $\hat{m}_k = \sum_{j=1}^q f_{jk} c_j$, and $\hat{b}_k = \sum_{j=1}^q f_{jk} r_j$, where $f_{jk} = h_j(\mathbf{x}_k) / \sum_{t=1}^q h_t(\mathbf{x}_k)$. To identify the connection weight $v_j = (l_j, c_j, r_j)$, the regular least squares method is firstly applied to identify its mode, c_j , by minimizing the mean square error $(\frac{1}{N} \sum_{k=1}^N (m_k - \hat{m}_k)^2)$, and a linear programming problem is then formulated to identify l_j and r_j , i.e.,

$$\text{Min} \quad \sum_{k=1}^N \sum_{j=1}^q f_{jk} \cdot (r_j - l_j), \quad (17)$$

$$\text{s.t.} \quad l_j \leq c_j \leq r_j, \quad j = 1, \dots, q, \quad (18)$$

$$\sum_{j=1}^q f_{jk} l_j \leq a_k, \quad k = 1, \dots, N, \quad (19)$$

$$\sum_{j=1}^q f_{jk} r_j \geq b_k, \quad k = 1, \dots, N. \quad (20)$$

The objective function of Eq. (17) is to minimize the fuzziness of the network outputs. Meanwhile, Eq. (18) constrains the feasibility of l_j and r_j ; and Eqs. (19) and (20) compose the inclusion condition proposed in Tanaka [24].

4.2. Generalization of FRBFN

A neural network may sometimes adapt itself so well to the training data that it loses its prediction capability for new input data, i.e., data supplied after the training period. Consequently, it is necessary to prevent the occurrence of over-fitting, or equivalently, to guarantee the generalization of the network. *Generalization* refers to constructing a model which performs a satisfactory mapping of the training data while still retaining the ability to approximate the outputs of any new input data.

The twin goals of accurately approximating the training data and generalizing the network tend to be conflicting objectives. A greater number of hidden units generally enhances the accuracy of the approximation, but inevitably leads to a poorer generalization of the network. Therefore, it is important to establish the appropriate number of hidden units in the FRBFN, i.e., the number of hidden units which ensures a high approximation accuracy while preserving an acceptable generalization for future unseen data. Specifying the optimal number of hidden units is achieved by means of the cross-validation technique. The principal of cross-validation is to divide the sample data into a construction sub-sample and a validation sub-sample. The construction sub-sample is used to train the network, and the validation sub-sample is used to test the generality of the network once it has been trained.

The steps of the cross-validation technique can be summarized as follows: (1) determine the number of hidden units (starting with a small number), (2) train the network with the training set data, (3) recall

the network trained in the previous step with the test set data, and record the mean square error produced by this network, (4) increase the number of hidden units and return to the first step, and (5) adopt the network which produces the minimum mean square error as the optimum configuration.

5. Fuzzy control charts

Construction of the fuzzy control charts requires a prior knowledge of the process mean. The process mean is estimated using a sample of products taken from the process when the process is thought to be in control. The quality outcome of each product within the sample is automatically obtained by supplying its dimensions to the fuzzy regression model (i.e., the FRBFN), as discussed in the previous section. Assuming that a sample of M fuzzy ratings $\hat{Y}_1, \dots, \hat{Y}_M$ has been obtained, the mean of the manufacturing process is estimated by

$$\bar{Y} = \frac{\sum_{k=1}^M \hat{Y}_k}{M}. \quad (21)$$

Let $\bar{Y} = (\bar{a}, \bar{m}, \bar{b})$. Then, through fuzzy arithmetic:

$$\bar{a} = \frac{\sum_{k=1}^M \hat{a}_k}{M}, \quad \bar{m} = \frac{\sum_{k=1}^M \hat{m}_k}{M}, \quad \bar{b} = \frac{\sum_{k=1}^M \hat{b}_k}{M}. \quad (22)$$

As the manufacturing process proceeds, sample data are drawn on a periodic basis and the mean of each sample is compared with $\bar{Y}$ to detect whether or not the process is still in control. As discussed in the following subsection, this comparison is performed on the basis of possibility and necessity measures.

5.1. Possibility measure and necessity measure

The principle of the present FPC methodology is to determine the degree of matching between the sample mean and the in-control process mean. If the degree of matching is high, then the process is considered to be in control. Specifically, let $\tilde{Y}$ be the mean of a sample drawn from the process and let this mean be an estimate of the current process level. It is then necessary to assess the matching degree of “ $\tilde{Y}$ is $\bar{Y}$ ” given the possibility distribution of $\tilde{Y}$. This assessment involves two related questions: (1) “Given the possibility distribution of $\tilde{Y}$, is it possible for $\tilde{Y}$ to be $\bar{Y}$?” and (2) “Given the possibility distribution of $\tilde{Y}$, is it necessary for $\tilde{Y}$ to be $\bar{Y}$?” The answer to the first question is referred to as a possibility measure, while the answer to the second is referred to as a necessity measure [29]. The answers to these two questions can also be interpreted as the degrees of truth that the process level is maintained as $\bar{Y}$ with the presence of $\tilde{Y}$.

Definition 1. The possibility measure of the variable $\tilde{Y}$ satisfying the condition “ $\tilde{Y}$ is $\bar{Y}$ ” is defined as [29]

$$\text{Pos}(\tilde{Y}|\bar{Y}) = \sup_{z \in U} [\min\{\mu_{\tilde{Y}}(z), \Pi_{\bar{Y}}(z)\}], \quad (23)$$

where $\Pi_{\bar{Y}}$ denotes the possibility distribution of $\bar{Y}$, U is the universe and z denotes the elements of U .

Definition 2. The necessity measure of the variable $\tilde{Y}$ satisfying the condition “ $\tilde{Y}$ is $\bar{Y}$ ” is defined as [29]

$$Nec(\bar{Y}|\tilde{Y}) = \inf_{z \in U} [\max\{\mu_{\tilde{Y}}(z), 1 - \Pi_{\tilde{Y}}(z)\}], \quad (24)$$

where $\Pi_{\tilde{Y}}$ denotes the possibility distribution of $\tilde{Y}$, U is the universe and z denotes the elements of U .

5.2. Construction of fuzzy control charts

Let $\bar{Y}$ be the mean of the in-control process. Once the manufacturing process has continued for a certain period, a further sample of products is collected. As before, the dimensions of each product are measured and supplied to the FRBFN in order to acquire the respective fuzzy quality ratings. The mean of the sample is computed as $\tilde{Y} = (\tilde{a}, \tilde{m}, \tilde{b})$, and serves as an estimate of the current process. For the process to remain in control, the possibility measure of the fuzzy sample mean, $\tilde{Y}$, must be greater than or equal to a threshold parameter α , whose value is specified in the range $0 < \alpha \leq 1$ by the user. To visualize the above condition on a control chart, the following property is defined:

Property 1. If $Pos(\bar{Y}|\tilde{Y}) \geq \alpha$, then $[\bar{Y}]_{\alpha} \cap [\tilde{Y}]_{\alpha} \neq \emptyset$, where $[\cdot]_{\alpha}$ denotes the α -level set.

Prade [18] has justified the possibility measure defined in Eq. (23) in terms of α -cuts and shown that it is equivalent to $\sup\{\alpha | [\bar{Y}]_{\alpha} \cap [\tilde{Y}]_{\alpha} \neq \emptyset\}$ [6]. This result proves Property 1.

The following example illustrates the possibility measures of five sample means.

Example 2. Assume that the in-control process mean is estimated by $\bar{Y} = (4, 5, 6)$. Five samples of observations are drawn from the process periodically, and their averages are calculated as $\tilde{Y}_1 = (4.5, 5.3, 5.8)$, $\tilde{Y}_2 = (3.6, 4.6, 5.6)$, $\tilde{Y}_3 = (2, 5, 8)$, $\tilde{Y}_4 = (2, 6, 7.5)$, and $\tilde{Y}_5 = (5.5, 6.5, 7.5)$, respectively. Let the threshold parameter for the possibility measure be specified as $\alpha = 0.8$. By Eq. (23), the possibility measures of these five samples are shown to be: $Pos(\bar{Y}|\tilde{Y}_1) = 0.83 > \alpha$, $Pos(\bar{Y}|\tilde{Y}_2) = 0.8 = \alpha$, $Pos(\bar{Y}|\tilde{Y}_3) = 1 > \alpha$, $Pos(\bar{Y}|\tilde{Y}_4) = 0.8 = \alpha$, and $Pos(\bar{Y}|\tilde{Y}_5) = 0.25 < \alpha$. To illustrate the relationship between the possibility measure and the α -level set of a fuzzy set as stated in Property 1, the α -level sets of the process mean and these five sample means are determined: $[\bar{Y}]_{\alpha} = [4.8, 5.2]$, $[\tilde{Y}_1]_{\alpha} = [5.14, 5.4]$, $[\tilde{Y}_2]_{\alpha} = [4.4, 4.8]$, $[\tilde{Y}_3]_{\alpha} = [4.4, 5.6]$, $[\tilde{Y}_4]_{\alpha} = [5.2, 6.3]$, and $[\tilde{Y}_5]_{\alpha} = [6.3, 6.7]$. As shown in Fig. 4, $Pos(\bar{Y}|\tilde{Y}_1) > \alpha$, and hence $[\bar{Y}]_{\alpha} \cap [\tilde{Y}_1]_{\alpha} \neq \emptyset$, and similarly, $[\bar{Y}]_{\alpha} \cap [\tilde{Y}_1]_{\alpha} \neq \emptyset$, $[\bar{Y}]_{\alpha} \cap [\tilde{Y}_3]_{\alpha} \neq \emptyset$, $[\bar{Y}]_{\alpha} \cap [\tilde{Y}_4]_{\alpha} \neq \emptyset$, and $[\bar{Y}]_{\alpha} \cap [\tilde{Y}_5]_{\alpha} = \emptyset$.

As illustrated by Example 2, the possibility measure can indicate the compatibility of the mode of the sample mean with that of the in-control process mean. Nevertheless, the degree of fuzziness of such a sample mean can still be far greater than is desirable, even when its possibility measure satisfies the mode compatibility requirement. This undesirable situation is evident in the sample means depicted in Fig. 4c and d. In such situations, a compensation step is performed using necessity measures.

A necessity measure not only assesses the necessity of the sample mean to conform to the in-control process mean, but also serves as a measure to indicate whether the fuzziness of a sample mean is too large compared to that of the in-control process mean. For the process to be in control, the necessity measure of the fuzzy sample mean $\tilde{Y}$ must be greater than or equal to a threshold β , whose value is specified in the range $0 < \beta \leq 1$ by the user. To visualize the necessity measure, the following property is defined.

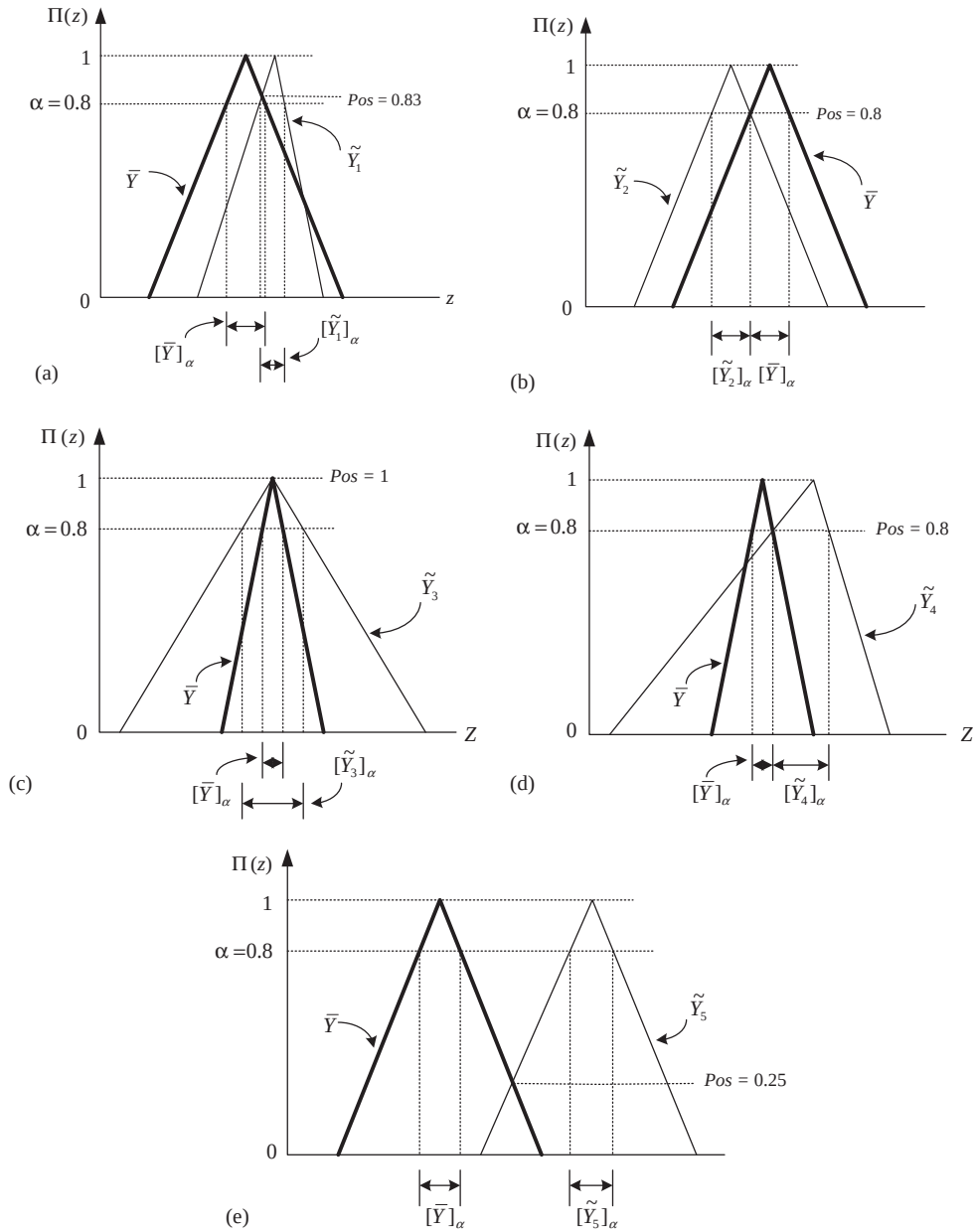


Fig. 4. (a) Possibility measure of Sample 1, $[\bar{Y}]_\alpha \cap [\tilde{Y}_1]_\alpha \neq \emptyset$; (b) possibility measure of Sample 2, $[\bar{Y}]_\alpha \cap [\tilde{Y}_2]_\alpha \neq \emptyset$; (c) possibility measure of Sample 3, $[\bar{Y}]_\alpha \cap [\tilde{Y}_3]_\alpha \neq \emptyset$; (d) possibility measure of Sample 4, $[\bar{Y}]_\alpha \cap [\tilde{Y}_3]_\alpha \neq \emptyset$; (e) possibility measure of Sample 5, $[\bar{Y}]_\alpha \cap [\tilde{Y}_5]_\alpha = \emptyset$.

Property 2. If $Nec(\bar{Y}|\tilde{Y}) \geq \beta$, then $[\bar{Y}]_\beta \supset [\tilde{Y}]_{1-\beta}$.

Property 2 was proved by Prade [18].

It can be readily proven that when $\tilde{Y} = \bar{Y}$, the necessity measure of $\tilde{Y}$ equals 0.5. Therefore, it is only meaningful to specify $0 < \beta < 0.5$.

Example 3. The five sample means in Example 2 are used again here to illustrate the necessity measure. Assume that the threshold value for the necessity measure is specified as $\beta = 0.3$. From Eq. (24), it is shown that the necessity measures of these five samples are: $Nec(\bar{Y}|\tilde{Y}_1) = 0.47 > \beta$, $Nec(\bar{Y}|\tilde{Y}_2) = 0.3 = \beta$, $Nec(\bar{Y}|\tilde{Y}_3) = 0.25 < \beta$, $Nec(\bar{Y}|\tilde{Y}_4) = 0 < \beta$, and $Nec(\bar{Y}|\tilde{Y}_5) = 0 < \beta$. The β -level set of the process mean is $[\bar{Y}]_\beta = [4.3, 5.7]$ and the $(1 - \beta)$ -level set of the five sample means are: $[\tilde{Y}_1]_{1-\beta} = [5.06, 5.45]$, $[\tilde{Y}_2]_{1-\beta} = [4.3, 4.9]$, $[\tilde{Y}_3]_{1-\beta} = [4.1, 5.9]$, $[\tilde{Y}_4]_{1-\beta} = [4.8, 6.45]$, and $[\tilde{Y}_5]_{1-\beta} = [6.2, 6.8]$. As shown in Fig. 5, $Nec(\bar{Y}|\tilde{Y}_1) \geq \beta$ and $Nec(\bar{Y}|\tilde{Y}_2) \geq \beta$. Hence $[\bar{Y}]_\beta \supset [\tilde{Y}_1]_{1-\beta}$ and $[\bar{Y}]_\beta \supset [\tilde{Y}_2]_{1-\beta}$, while the other sample means indicate $[\bar{Y}]_\beta \not\supset [\tilde{Y}_3]_{1-\beta}$, $[\bar{Y}]_\beta \not\supset [\tilde{Y}_4]_{1-\beta}$, and $[\bar{Y}]_\beta \not\supset [\tilde{Y}_5]_{1-\beta}$.

In accordance with the above discussions, the out-of-control conditions for the sample means are formulated as follows.

Remark 1. Given the in-control process mean $\bar{Y}$, a fuzzy process is considered to be out-of-control if, for a sample mean $\tilde{Y}$, either $[\bar{Y}]_\alpha \cap [\tilde{Y}]_\alpha = \emptyset$ or $[\bar{Y}]_\beta \not\supset [\tilde{Y}]_{1-\beta}$, where α and β are thresholds specified by the user.

Based on Remark 1, Fig. 6 presents two fuzzy control charts on which the five sample means of Examples 2 and 3 are plotted. In Fig. 6a, the band bounded by the interval $[L_\alpha, U_\alpha]$ represents the α -level set of $\bar{Y}$ and the vertical line segments represent the α -level sets of the five sample means. Similarly, in Fig. 6b, the band bounded by the interval $[L_\beta, U_\beta]$ represents the β -level set of $\bar{Y}$ and the vertical line segments represent the $(1 - \beta)$ -level sets of the five sample means. From Fig. 6a, it is observed that all samples other than Sample 5 intersect with the band. This indicates that the possibility measures of Samples 1–4 are greater than or equal to the threshold α . Meanwhile, it is noted in Fig. 6b that only Samples 1 and 2 are included in the band. This implies that only these two samples have a necessity measure greater than the threshold β . Therefore, it can be concluded that Samples 1 and 2 conform to the process, while the remaining samples do not.

6. Discussion and concluding remarks

This paper has presented a novel methodology for FPC whose purpose is to monitor a process with fuzzy outcomes represented by fuzzy numbers. Two fuzzy control charts have been constructed to directly monitor the fuzzy outcomes in order to establish whether or not the process is in control. The out-of-control conditions of these control charts are formulated in accordance with possibility theory. The joint application of these two control charts permits both the central tendency and the fuzziness of the process to be monitored.

It is noted that the thresholds of the possibility and necessity measures (i.e., α and β) play important roles in determining the out-of-control conditions, which justify the conformance or otherwise of samples to the process. A similar concern was noted in the membership approach proposed by Raz and Wang [19], in which the user was required to specify the multiplier of the mean deviation utilized to establish the control limits of the control charts. Raz and Wang [19] proposed the use of Monte–Carlo simulation

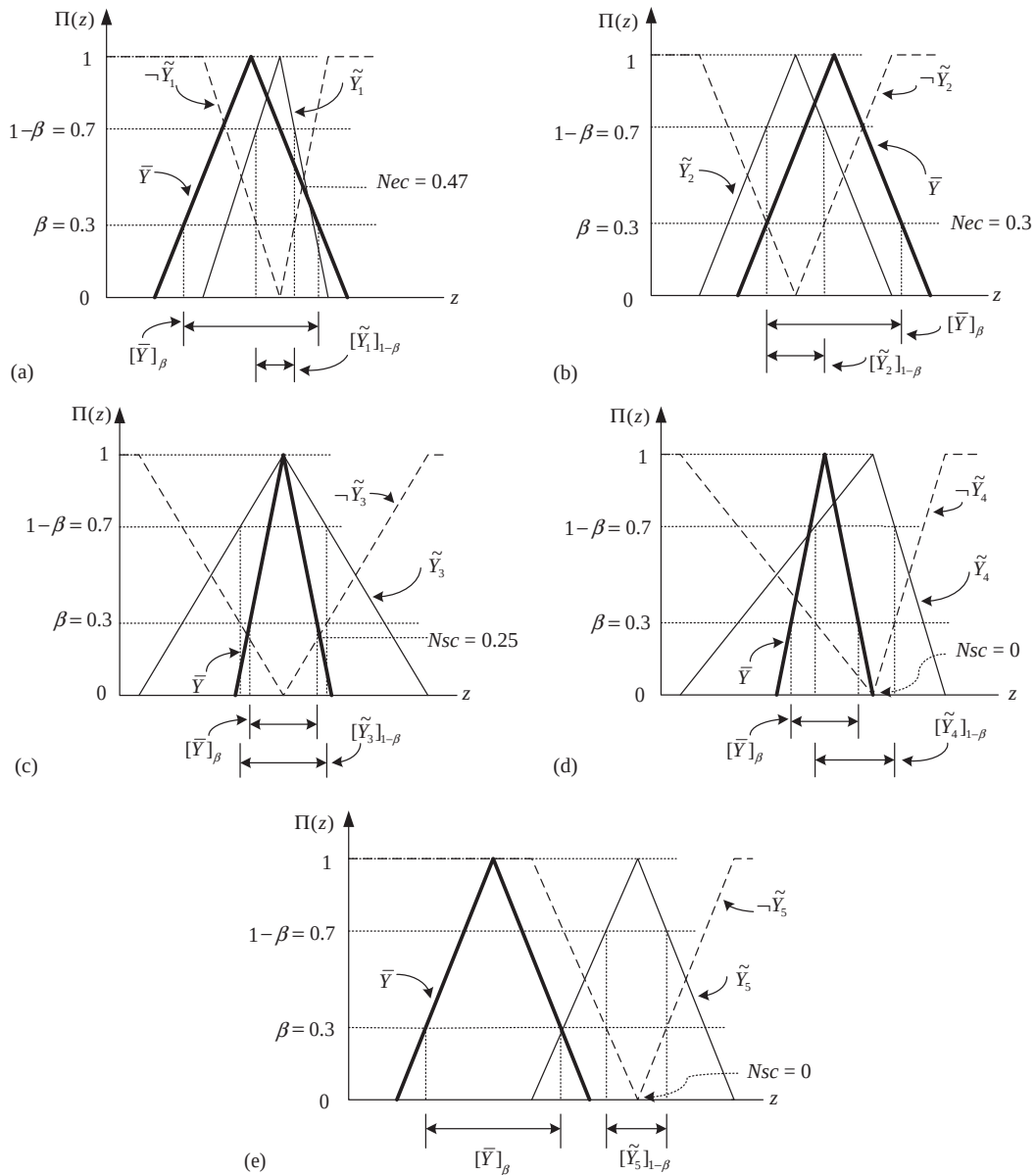


Fig. 5. (a) Necessity measure of Sample 1, $[\bar{Y}]_\beta \supset [\tilde{Y}_1]_{1-\beta}$; (b) necessity measure of Sample 2, $[\bar{Y}]_\beta \supset [\tilde{Y}_2]_{1-\beta}$; (c) necessity measure of Sample 3, $[\bar{Y}]_\beta \not\supset [\tilde{Y}_3]_{1-\beta}$; (d) necessity measure of Sample 4, $[\bar{Y}]_\beta \not\supset [\tilde{Y}_4]_{1-\beta}$; (e) necessity measure of Sample 5, $[\bar{Y}]_\beta \not\supset [\tilde{Y}_5]_{1-\beta}$.

to establish the value of the multiplier which would yield a pre-specified probability of Type I error. The thresholds α and β in the control charts of the present study can also be established based on a pre-specified probability of Type I error. If such an approach is adopted, empirical trials can be conducted in a trial-and-error manner to obtain appropriate threshold values. In these trials, many samples are drawn

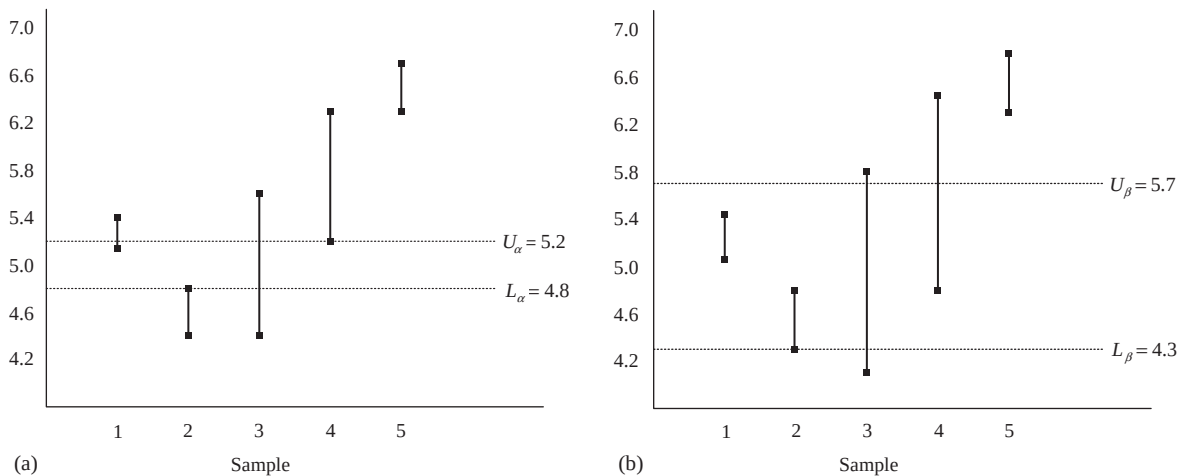


Fig. 6. (a) Fuzzy control chart obtained by possibility measure, (b) fuzzy control chart obtained by necessity measure.

from the process when it is thought to be in control. Initial values of α and β are then assigned by the user and the control charts are constructed using the approach presented in Section 5. If the fraction of out-of-control samples in the control chart of Fig. 6a is found to be greater than the pre-specified probability of Type I error, then the value of α is decreased, otherwise, its value is increased. The control chart is then re-constructed with the new α value and the fraction of out-of-control samples calculated once again. This iterative procedure is repeated until the fraction of out-of-control samples is close to the pre-specified probability of Type I error. Similarly, if the fraction of out-of-control samples in the control chart of Fig. 6b is greater than the pre-specified probability of Type I error, then β is decreased, otherwise, its value is increased.

The quality assessment based on the inspectors' rating scores proposed in this study provides the potential of using the Choquet integral [4] to construct the possibility and necessity measures used for formulating the control charts. Klir et al. [15] presented a data-driven method for constructing fuzzy measures using the Choquet integral. Consider the case of a set of quality attributes, a function assigning a quality value to each attribute of a given object, and an unknown fuzzy measure specifying the importance of each subset of the attributes. With overall assessments of the objects given by experts, the Choquet integral can be used to solve the fuzzy measure. The possibility and necessity measures used in this study can be constructed using a similar technique. The use of the Choquet integral in fuzzy process control will be investigated in a future study.

In future research, further rules will also be developed for detecting the variation of a process in order to increase the sensitivity of the fuzzy control charts to small process shifts or to other unusual patterns such that users may respond more rapidly to special causes.

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